

Similarity statement

The MP dataset includes 41 different silicon-only structures, however, many of them are very high-density (high coordination number) or crystalline (all 4-fold). There were no cases of wide coordination number distribution, as seen in liquid Si – however, we found 5 unique a-Si structures (with 100 atoms each) with a mix of slightly higher and lower coordination numbers, providing information about a-Si. Based on a UMAP analysis, the closest structures in the training set are mp-1244971, mp-1245242 and mp-1245041. To help with visualization, we provide `amorphous-silicon.json` on chemiscope.org.

Performance summary

The model performs reasonably well for the description of the melt–quench process, leading to good-quality a-Si structures, albeit markedly underestimating the vitrification temperature and the excess enthalpy and overestimating the number of 5-fold coordinated atoms compared to a domain-specific ML potential.

A.3 Amorphous carbon

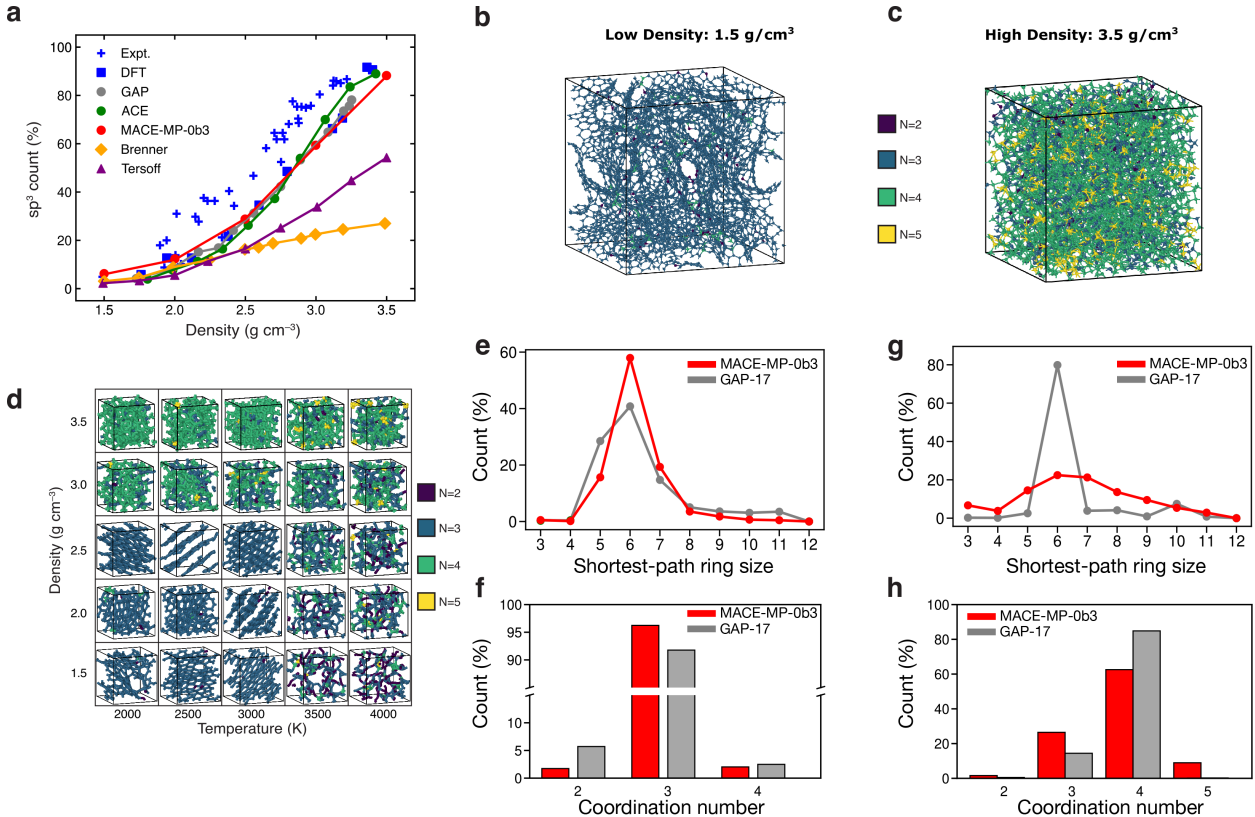


Figure S4: **Amorphous carbon.** (a) Count of sp^3 (fourfold coordinated) carbon atoms in melt-quenched carbon structures as a function of density. The results obtained with the MACE-MP-0b3 model are compared to computational and experimental data compiled in Ref. (173) and references therein, as well as Refs. (174) and (175). (b–c) 4,096-atom structures generated using the MACE-MP-0b3 potential. (d) Results from 25×200 atom graphitisation simulations spanning relevant temperature and density ranges, similar to Ref. (176). The structures in panels (b–d) are colour-coded according to coordination numbers as indicated in the legends. (e) Shortest-path ring size count for 4,096-atom low-density structures as determined using *matscopy* (172). (f) Coordination number count for 4,096-atom low-density structures. (g–h) Same as panels (e) and (f) but for high-density structures. The results in panels (e–h) are shown for a C-GAP-17-driven simulation (gray) and for a comparable simulation driven by the MACE-MP-0b3 potential (red).

A.3.1 Melt-quench simulations

Carbon forms many different crystalline and amorphous modifications. The structural diversity of amorphous carbon (a-C), characterised by the simultaneous presence of three-fold coordinated (sp^2) and four-fold coordinated carbon atoms (sp^3), makes it a challenging system for both classical and ML force fields (177, 178). The correct description of its growth mechanism has been among the early successes of ML-driven materials modeling (179).

We assess the accuracy of the MACE-MP-0 model in reproducing the structural complexity of amorphous phases by plotting the concentration of four-fold coordinated atoms (sp^3) as a function of density in Fig. S4. To generate amorphous structures with a given density with the MACE-MP-0 model, we perform melt-quench simulations. We start by melting diamond structures at a given density by running NVT simulations at 8000 K for 3 ps. We then perform a fast quench, reducing the temperature from 8000 K to 300 K at a cooling rate of 1000 K/ps. Finally, we optimize the geometry with LFBGS to obtain the final structure and

determine the count of sp^3 atoms using a bond-length cutoff of 1.85 Å. We observe in Fig. S4 that the MACE-MP-0 model predictions reproduce the trend observed in both the DFT (174) and the experimental data extracted from (173). We also see good agreement with results of quenches using the carbon ACE reported by Qamar et al. in Ref. (175), and using C-GAP-17 reported in Ref. (174), noting that both potentials had been specifically trained on large carbon datasets.

A.3.2 Graphitisation

MACE-MP-0 was used for two annealing runs for simulation cells containing 4096 atoms at low density (1.5 g/cm³) and high density (3.5 g/cm³), respectively. The low-density simulation was run at 2000 K and the high-density simulation was run at 4000 K. Additionally, 25×200 atom annealing runs spanning from 2000 to 4000 K and 1.5 to 3.5 g/cm³ were conducted to explore a finer grid of parameters. Both 4,096-atom structures were compared to structures generated using C-GAP-17 using the same protocol, which was also used recently for an “on-the-fly” generated GAP potential (176). The protocol has two stages. The goal of stage I is to prepare the starting configuration for the annealing in stage II, and it begins with a random structure with a hard-sphere constraint of $r_{\min} \geq 1$ Å and equilibrating it at 9000 K for 40 ps, followed by cooling to 5500 K over 40 ps and subsequent quenching to 300 K over 10 ps. The structures are then held at 300 K over 50 ps before being rapidly heated up to the annealing temperature over 10 ps. This concludes stage I. In stage II, the structure is annealed at 2000 K or 4000 K for 350 ps using a time step of 1 fs. We used C-GAP-17 to perform stage I because MACE-MP-0 was found to be unstable at 9000 K.

For the low-density case, the structures generated by MACE-MP-0 in annealing simulations agree with the predictions of the established C-GAP-17 model in terms of overall trends (Figs. S4b and S4d). For more detailed insight, Fig. S4e shows the shortest-path ring distribution for the low-density structure, indicating that MACE-MP-0 predicts a greater number of 6-membered rings and fewer large rings for the low-density structure compared to C-GAP-17 (thus suggesting a higher degree of crystallinity in the MACE-MP-0 prediction) – this might be correlated with the higher relative count of graphite versus amorphous-like structures in the training dataset, although further analysis is required.

Figure S4g shows poor agreement between MACE-MP-0 and C-GAP-17 in terms of ring count for the high-density structure. Whereas annealing with C-GAP-17 led to a partly crystallised structure under the conditions chosen (indicated by the large abundance of 6-membered rings, as are found in the diamond structure), the MACE-MP-0b3 model gave rise to a highly disordered structure – this is evident from a visual inspection of Fig. S4c, and reflected in the ring-count plot in Fig. S4g showing a notable number of (presumably strained) 3-membered rings as well as larger ring sizes of > 7 . In other words, the MACE-MP-0b3 appears to fail to describe the crystallisation in this case (we note that a partially ordered structure was obtained in a 200-atom simulation at 3000 K but not at 4000 K at the same density; Fig. S4d). Finally, Figs. S4f and S4h show the coordination number for both 4,096 atom structures. MACE-MP-0b3 leads to more sp^2 environments and fewer sp and sp^3 environments compared to C-GAP-17 (again likely consistent with a higher degree of crystallinity, as suggested by the ring count in Fig. S4e). In the high-density 4,096-atom structure obtained with MACE-MP-0b3, a notable amount of 5-fold coordinated atoms are observed, on the order of about 10% – a behaviour that would not be expected in a simulation of carbon at diamond-like density.

We note that these issues are not expected to be a fundamental shortcoming of the MACE architecture: we show that training on a wider-ranging dataset with more crystal structures like the Alexandria dataset (which notably has no amorphous structures) results in a model, MACE-MPA-0, that describes the structure of amorphous carbon in better agreement with C-GAP-17 at high and low density, as shown in the section dedicated to the MACE-MPA-0 model (A.33).

Similarity statement

The MP dataset contains 89 different all-carbon structures, most of which correspond to diamond and related stacking polytypes (lonsdaleite-like and more complex ones, all purely sp^3 -bonded), as well as graphite in various forms. These structures include a range of mixed configurations with sp^2/sp^3 coexistence – the latter is expected to be critical for a correct description of a-C. The dataset also contains a number of compressed and defective fullerene configurations, with one of those cells containing sp -, sp^2 -, and sp^3 -like environments, and a few hypothetical allotropes (notably “T-carbon” and a cubane-motif-based form, representing 3- and

4-membered shortest-path rings, respectively). In essence, the dataset does contain relevant carbon environments but does not contain a significant share of highly disordered carbon configurations. Based on UMAP analysis, we find that the closest structures in the training set are mp-568028 and mp-568806. We provide `amorphous_carbon.json` to help visualize the interactive UMAP on chemiscope.org.

Performance summary

The MACE-MP-0 model correctly captures the sp^3 content as a function of density in melt-quench simulations. Detailed analysis of long-annealing simulations shows qualitative agreement with a purpose-trained ML force field for low-density a-C graphitization and poor agreement for high-density annealing.

A.4 Ceria nanoparticles

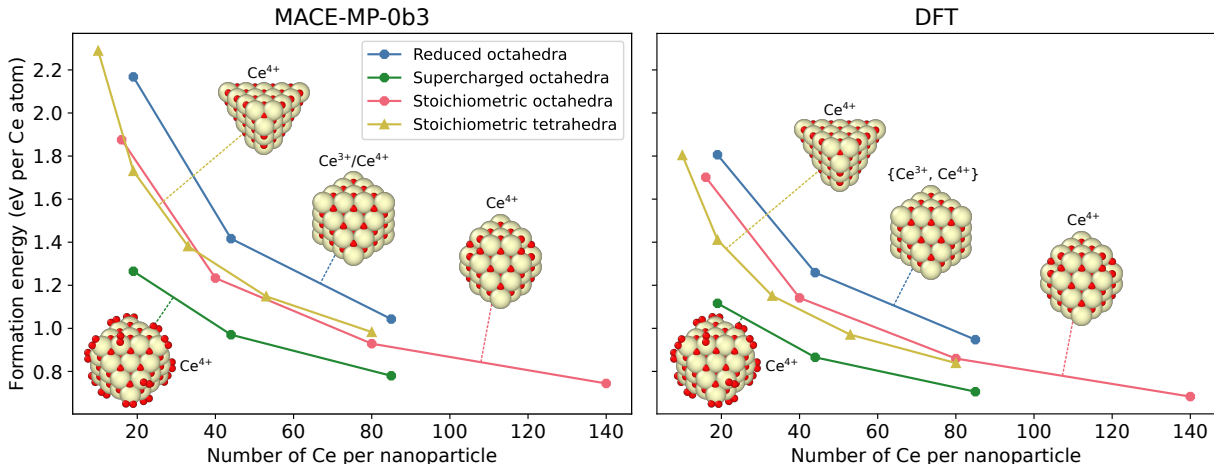


Figure S5: Size-dependent formation energies for different shapes and stoichiometries of ceria nanoparticles (NPs). Left panel: Results calculated with the MACE-MP-0 model, which was not trained on any data from ceria surfaces or NPs. Right panel: Independent validation data from PBE+U calculations. The NP images are just illustrations; the optimized structures from MACE-MP and DFT are compared in more detail in Fig. S6. The formation energy is calculated with respect to stoichiometric bulk CeO_2 (and gas-phase O_2 molecules as needed).

Cerium oxide (ceria, CeO_2) is a reducible metal oxide with intriguing chemical and physical properties, and important technological applications especially for nanostructured ceria. Experiments in the literature have for example shown that the oxygen storage capacity (OSC) of ceria at the nanoscale is strongly shape- and size-dependent.

Behind the versatile usage of ceria lies one overriding feature, namely, its exceptional reduction-oxidation (redox) properties enabled by the duality of the cerium ion ($\text{Ce}^{4+} \longleftrightarrow \text{Ce}^{3+}$). It is generally a formidable task to try to mimic interactions, structure and energetics simultaneously for a compound like ceria without having access to explicit electrons. In earlier work (180), we constructed a reactive interaction model using the ReaxFF modelling framework (181) with the aim of handling stoichiometric and partially reduced ceria bulk, surfaces, and nanoparticles (NPs). The model was based on a training set of DFT calculations for a large number of ceria systems in various forms and configurations (bulk, clusters, surface systems; stoichiometric as well as reduced systems). With some exceptions that model performed very well. In the present study, instead, our forcefield is MACE-MP-0, where the content of ceria in the training set is only bulk structures, namely exactly 18 bulk polymorphs (stoichiometric or partially reduced).

Here we assess the ability of the MACE-MP-0 model to describe small ceria nanoparticles of different shapes, sizes and reduction degrees without the training ever including any ceria NPs or surfaces. The optimized NP structures/shapes as well as their formation energies (with respect to stoichiometric bulk CeO_2 and gas-phase O_2 as needed) will be assessed.

The left panel in Fig. S5 shows the MACE-MP-0 results for optimized particles up to 140 formula units: stoichiometric tetrahedra, stoichiometric truncated octahedra, and perfect octahedra (which are partially reduced by virtue of their shapes). The bottommost curve pertains to “supercharged” NPs, i.e. perfect octahedra that are decorated with oxygen molecules. The right panel shows the corresponding results from independent reference calculations at the DFT (PBE+U) level, taken from Refs. (180, 182)

The agreement between the energetics in the two panels in Fig. S5 is good overall, which is satisfying. However, we note that as far as structures are concerned, the MACE-MP-0 model (leftmost panel of Fig. S6) is unable to distinguish between Ce^{3+} and Ce^{4+} ions, both of which should in fact be present in a partially reduced perfectly octahedral ceria nanoparticle. This deficiency of the MACE-MP-0 structure is evident from a comparison with the independent electronic PBE+U calculations in the rightmost panel of Fig. S5,